

# ENVS 193DS Spring Final

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## Set Up

```
# reading in packages
library(tidyverse)
library(janitor)
library(DHARMA)
library(here)
library(MuMIn)
library(ggeffects)
# reading in nest data
nest <- read.csv(here("data/nest_data_final.csv"))
# reading in my personal data
my_data <- read.csv(here("data/personal_data.csv"))
```

## Problem 1. Research writing

### a. Transparent statistical methods

In part one my co-workers are using a generalized linear model test. This is seen through the response variable, whether the American Avocet utilized the habitat (San Francisco Bay), being binary data (yes or no). Additionally the binary response variable is being linked to a continuous variable, the distance from the water edge, further signalling the use of a generalized linear model.

In part two my co-workers are using a chi-squared test. This is seen through the analysis testing whether there is a relationship between two categorical variables: bird species and habitat.

## b. Suggestions for rewriting

- **Part 1:** The data indicates that the distance to the water's edge significantly influences whether American Avocets utilizes a particular microhabitat. This means an avocet's likelihood of occupying an area changes predictably depending on how close or far the area is from the edge of the water.
- **Part 2:** We found clear evidence that different shorebird species (American Avocet, Forter's Tern, Black-necked Stilt) prefer distinct types of habitats (managed pond, non-tidal marsh, salt pond, tidal marsh, or tidal mudflat) over others in the San Francisco Bay.

## c. Further analysis

- **Additional Variable:** An additional variable that could influence American Avocet microhabitat use is water depth.
- **Variable Type:** Water depth is a continuous variable that can be measured in centimeters (cm).
- **Why?:** Water depth may influence habitat use because American Avocets are wading shorebirds that require shallow water to effectively find food. Due to this, analyzing water depth could compliment the distance to water edge data and provide a more comprehensive understanding to the specific ecosystem conditions that impact shorebird habitat use.

## Problem 2. Data analysis

### a. Clean and wrangle

```
# creating clean object
nests_clean <- nest |>
# cleaning names
clean_names() |>
# changing height_cm to be in meters
mutate(height_m = height_cm / 100) |>
# selecting for desired columns
select(height_m,
       case_control,
       ant_species) |>
# filtering for multiple values using %in%
filter(ant_species %in% c("AB", "RRB", "BBR")) |>
```

```

# full scientific names in new column
mutate(species_name = fct_recode(as.factor(ant_species),
  "C. sjostedti" = "AB",
  "C. mimosae"   = "RRB",
  "C. nigriceps" = "BBR"
)) |>
# reordering columns in desired order
mutate(species_name = fct_relevel(species_name,
  "C. sjostedti",
  "C. mimosae",
  "C. nigriceps")) |>

# arranging by species_name
arrange(species_name)
# displaying 10 random rows
slice_sample(nests_clean, n = 10)

```

|    | height_m | case_control | ant_species | species_name |
|----|----------|--------------|-------------|--------------|
| 1  | 2.160    | 1            | RRB         | C. mimosae   |
| 2  | 1.030    | 0            | RRB         | C. mimosae   |
| 3  | 2.365    | 1            | RRB         | C. mimosae   |
| 4  | 3.530    | 0            | RRB         | C. mimosae   |
| 5  | 1.300    | 0            | RRB         | C. mimosae   |
| 6  | 1.950    | 1            | RRB         | C. mimosae   |
| 7  | 2.430    | 0            | RRB         | C. mimosae   |
| 8  | 1.320    | 0            | BBR         | C. nigriceps |
| 9  | 2.125    | 0            | RRB         | C. mimosae   |
| 10 | 1.555    | 0            | RRB         | C. mimosae   |

## b. Response variable

The 1s and 0s in this dataset, specifically seen in the `case_control` column, signal whether a whistling thorn tree has a bird nest. A 1 represents the presence of a bird nest, therefore signalling the tree as ‘used’, and a 0 represents the absence of a bird nest, therefore signalling the tree as ‘available’.

## c. Purpose of study

**Tree Height (m):** Tree height is a continuous variable measured in meters (m). As tree height increases the probability of nest occurrence would increase likely due to increased protection from predators (being higher off the ground) and structural advantages to taller trees (such as

increased support). The relationship between tree height and nesting was found in the results section, paragraph one. Reasoning for why this may be helpful was inferred by me based on common biological strategies.

**Acacia Ant Species:** Acacia ant species is a categorical variable that represents distinct nonnumerical groups based on the identity of the ant symbiont occupying the tree. As the presence of aggressive ant species (*C. mimosae* or *C. nigriceps*) increases, the probability of nest occurrence would increase because these specific ants defend the host trees against predators without attacking the birds or their nestlings. Conversely, trees occupied by less aggressive ants (*C. sjostedti* or *T. penzigi*) or no ants at all would see a decrease in the probability of nest occurrence because they offer minimal protection from nest predation. I found this information from the results section in paragraph 3 and the discussion section in paragraph 1.

#### d. Table of models

| Model Number   | Predictors Present       | Interaction Term? | Description                                                                                                                   |
|----------------|--------------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Model 1</b> | Tree height, Ant species | No                | Examines the independent main effects of tree height and ant species on the probability of nest occurrence.                   |
| <b>Model 2</b> | Tree height, Ant species | Yes               | Examines whether the effect of tree height on the probability of nest occurrence depends on the specific ant species present. |

#### e. Run the models

#### f. Select the best model

```
# calculating AIC from models
AICc(
  model1,
  model2
) |>
```

```
# rearranged to lowest to highest values
arrange(AICc)
```

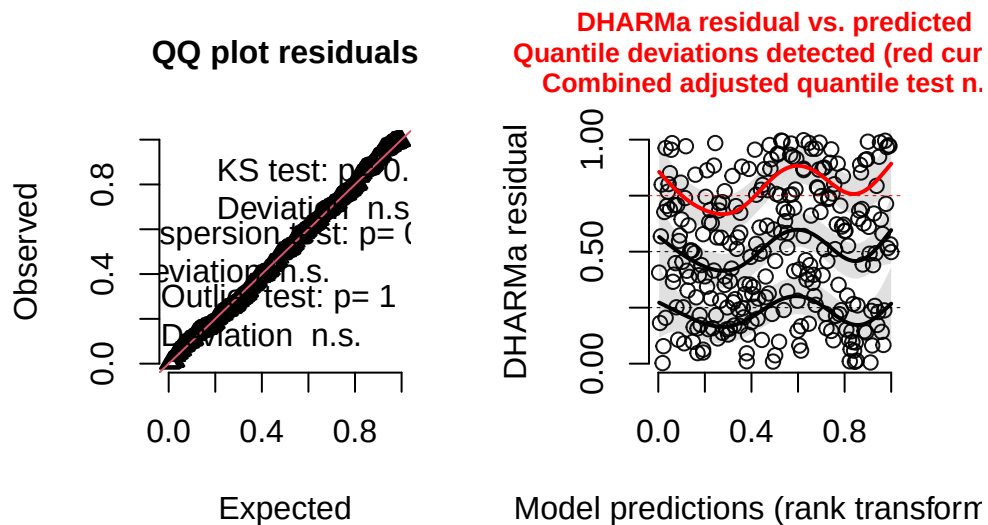
```
      df    AICc
model2  6 238.1236
model1  4 258.8940
```

The best model as determined by Akaike's Information Criterion (AIC) included the effects of tree height and acacia ant species, as well as the interaction between the two, to predict the probability of nest occurrence.

### g. Check the diagnostics

```
# checking diagnostics with DHARMA package
simulated_res <- simulateResiduals(fittedModel = model2)
# plot the DHARMA diagnostics
plot(simulated_res)
```

#### DHARMA residual



- **QQ Plot Residuals:** Points largely follow the reference line, indicating normality of residuals.

- **Residuals vs. Predicted:** The points appear randomly distributed without a distinct pattern, and the quantile lines are relatively flat, indicating the model fits the data.

## h. Visualize the model predictions

```
# defining colors
my_ant_colors <- c(
  "C. sjostedti" = "orchid3",
  "C. mimosae"   = "palegreen4",
  "C. nigriceps" = "skyblue2"
)
# creating object, preds, to store model predictions
preds <- ggpredict(model2, terms = c("height_m", "species_name"))
```

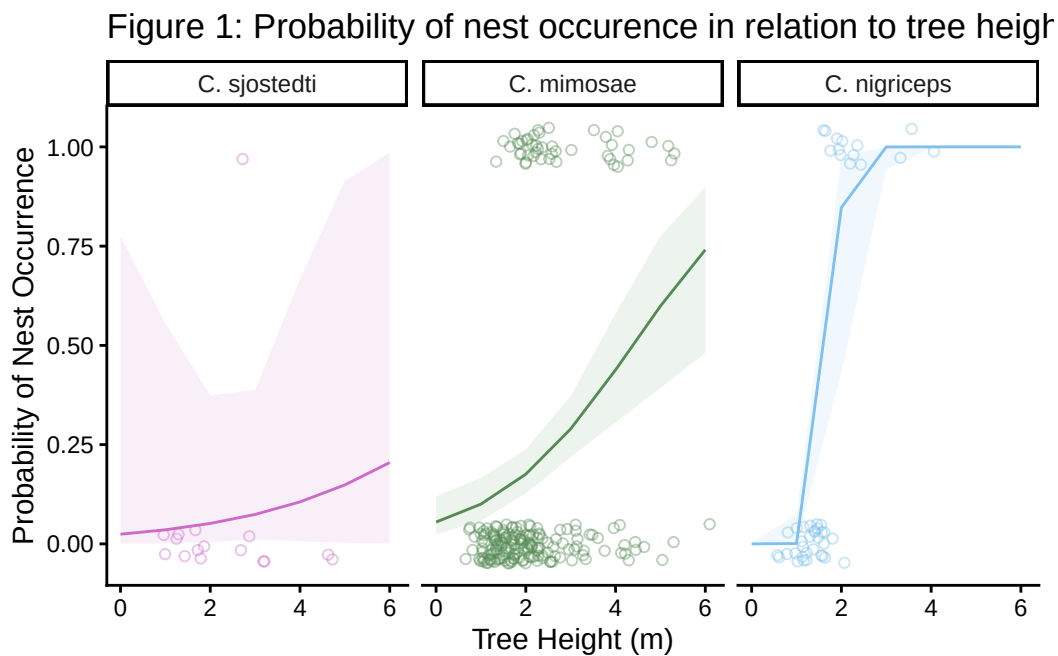
Data were 'prettified'. Consider using `terms="height\_m [all]"` to get smooth plots.

```
# changing the preds "group" column name to "species_name"
colnames(preds)[colnames(preds) == "group"] <- "species_name"
# base layer: ggplot
ggplot(mapping = aes(x = height_m, # x-axis height
                     y = case_control)) + # y-axis probability
# first layer: geom_jitter
geom_jitter(data = nests_clean,
            aes(color = species_name), # coloring by species name
            height = 0.05,
            width = 0.2,
            shape = 21,
            alpha = 0.4) +
# second layer: geom_line on preds data
geom_line(data = preds, mapping = aes(x = x,
                                       y = predicted,
                                       color = species_name)) +
# third layer: geom_ribbon to show 95% CI
geom_ribbon(data = preds, mapping = aes(x = x,
                                       y = predicted,
                                       ymin = conf.low,
                                       ymax = conf.high,
                                       fill = species_name),
           alpha = 0.1,
```

```

        color = NA) +
# making colors match my_ant_colors
scale_color_manual(values = my_ant_colors) +
scale_fill_manual(values = my_ant_colors) +
# faceting by species name
facet_wrap(~ species_name) +
# adding a theme
theme_classic() +
# removing legend
theme(legend.position = "none") +
# adding axes titles and a title for figure
labs(x = "Tree Height (m)",
      y = "Probability of Nest Occurrence",
      title = "Figure 1: Probability of nest occurrence in relation to tree height and ant sp

```



**i. Write a caption for your figure**

**Figure 1: Probability of nest occurrence in relation to tree height and ant species.** This plot shows the probability of nest occurrence for three bird species (gray-headed sparrow, gray-capped social weaver, superb starling) across varying tree heights (m) faceted by ant species (*C. sjostedti*, *C. mimosae*, and *C. nigriceps*). The solid lines depict mean model

predictions, with the shaded ribbons around the lines showing 95% confidence intervals. The open circles represent individual observations (1 = nest present, 0 = nest absent). Colors correspond to the ant species present at the tree: *C. sjostedti* (pink), *C. mimosae* (green), and *C. nigriceps* (blue). **Data source:** Lujan, E., Nielsen, R., Short, Z., Wicks, S., Nderitu Watetu, W., Malingati Khasoha, L., Palmer, T. M., Goheen, J. R., & Alston, J. M. (2023). Data, code, and metadata for: Symbiotic acacia ants drive nesting behavior by birds in an African savanna [Data set]. Zenodo. <https://doi.org/10.5281/zenodo.8373322>

## j. Calculate model predictions

```
predictions_at_6m <- ggpredict(model2,
                                terms = c("height_m [6]", "species_name"))

# This output will show you exactly the probability for 6m for all three species
print(predictions_at_6m)
```

# Predicted probabilities of case\_control

species\_name: *C. sjostedti*

| height_m | Predicted | 95% CI     |
|----------|-----------|------------|
| 6        | 0.20      | 0.00, 0.99 |

species\_name: *C. mimosae*

| height_m | Predicted | 95% CI     |
|----------|-----------|------------|
| 6        | 0.74      | 0.48, 0.90 |

species\_name: *C. nigriceps*

| height_m | Predicted | 95% CI     |
|----------|-----------|------------|
| 6        | 1.00      | 1.00, 1.00 |

## k. Interpret your results

As shown in **Figure 1**, the probability of bird nest occurrence increases significantly with tree height for trees inhabited by the aggressive ant species *C. mimosae* and *C. nigriceps*,



reaching a predicted probability of 0.74 and 1.00 respectively at a height of 6 m. In contrast, the probability remains low (0.20) for trees occupied by the less aggressive ant species, *C. sjostedti*, highlighting a clear preference for trees defended by more aggressive ant symbionts. This pattern aligns with the study's findings that birds prioritize nesting in trees defended by aggressive ants to potentially reduce nest predation risk. Furthermore, the stronger association seen with *C. nigriceps* may be due to the species' tendency to prune apical buds, creating a denser more concealed canopy that provides birds with additional physical protection from predators.